MediLink Engineering Management Report

Part A — Team & Development Workflow

1. Team Responsibilities

Team Member	Role	Core Responsibilities
Michael Girgis	Engineering Manager & Developer	Process flow, risk management, impediment removal, developer coaching.
Matthew Halliburton	Scrum Master & Developer	Asynchronous updates tracking, WIP enforcement, metric dashboards.
Ricardo Morales Lopez	Product Owner & Developer	Backlog grooming, UI mockup approval, form field validation rules.
Cyril Szekiel Costes	Developer	Simple client-side form layout, HTML structure, CSS styling, basic JS client validation.

2. Scoped Backlog (Simple Form Increment — 8 Work Items)

ID	Title	Type	Priority	Owner	Acceptance Criteria
MED-201	Institution Dropdown Form Component	Feature	High	Cyril Costes	Static HTML <select> dropdown populated with hardcoded sample clinics (e.g., Montreal Medical, Downtown Dental).

MED-202	Client Personal Info Form Fields	Feature	High	Cyril Costes	Form inputs for Age (number), Gender (select), Weight (number), and Height (number).
MED-203	Client-Side Form Validation Script	Technical Work	High	Ricardo Lopez	Simple JS checks, preventing form submission if age <= 0, negative weight/height, or unselected institution.
MED-204	Form Responsive CSS & Layout Styling	Maintenance	Medium	Michael Girgis	Form centers cleanly on screen and remains readable on desktop and mobile viewports.
MED-205	Inline Form Validation Error Messaging UI	Defect	High	Cyril Costes	Red inline text hints display under invalid fields instead of generic browser alert popups.
MED-206	Form Submission Confirmation View State	Feature	Medium	Matthew Halliburton	Submitting a valid form hides input fields and displays a simple 'Form Submitted Successfully' confirmation card.

MED-207	Local Storage Client Intake Draft Saver	Technical Work	Low	Ricardo Lopez	Draft input values auto-save to browser localStorage so values persist upon page refresh.
MED-208	WCAG Accessibility & Form Labeling Refactor	Maintenance	Low	Michael Girgis	Add explicit <label> associations and ARIA attributes for screen reader accessibility compliance.

3. Workflow Rules & Definition of Done

Category	**Rule / Policy**
Workflow Steps	Backlog -> Ready -> In Progress -> Code Review -> Verification -> Done
WIP Limit	<= 1 active item per developer across In Progress, Code Review, and Verification.
Definition of Done (DoD)	HTML/CSS/JS code passes local browser sanity check -> Zero console errors -> Peer reviewed via GitHub PR -> PO visual approval.

Part B — Plan & Coordinate

1. SMART Objectives

Team Member	**SMART Objective**
Michael Girgis	Conduct 1 weekly coaching session to resolve developer bottlenecks with zero direct code hand-offs.
Matthew Halliburton	Maintain daily WIP limits, keeping PR review turnaround under 12 hours for the sprint.

Ricardo Morales Lopez	Refine 100% of backlog items with complete acceptance criteria before sprint start.
Cyril Szekiel Costes	Ensure 100% of form inputs satisfy client-side boundary validation before opening PRs.

2. Communication Protocols

Mechanism	**Format & Policy**
Daily Standup	Asynchronous team chat by 9:30 AM daily using template: [Completed] | [Planned] | [Blockers/Assistance Needed].
Synchronous Switches	1. Active production/preview outage. 2. PR review thread exceeding 5 comment exchanges.

Part C — Workflow Observations

Observation	**Evidence**	**Possible Explanation**	**Proposed Response**
1. Excessive Review Wait Time	PRs sat in review for >36 hours while devs pulled new tasks.	Developers prioritized writing new code over reviewing peer code.	Enforce WIP limit: No dev pulls new work if a PR awaits review.
2. Incomplete Requirements	MED-202 lacked min/max boundary definitions for age and weight inputs.	Rushed backlog grooming by Product Owner.	Enforce a strict 'Definition of Ready' checklist before sprint entry.

Part D — Production Incident & Root Cause Analysis

Phase	**Finding / Investigation Result**
Incident Description	Client intake form allowed submission of negative numbers for age and weight in production static preview.
Symptom	Form submitted successfully with invalid data like Age = -5 or Weight = -150.
Immediate Cause	HTML <input type='number'> elements lacked min attributes and JavaScript submit handler skipped input sanity checks.

Contributing Factors	No client-side unit test or DOM verification step was performed prior to merging.
Underlying Causes	Lack of explicit validation rules in the Definition of Done checklist for simple frontend forms.
Corrective Action	Add min='1' attributes to HTML inputs and enforce JS e.preventDefault() if validation fails.
Preventive Action	Include manual form boundary testing in the Definition of Done checklist.

Part E — Risk Management Register

Risk Type	Description	Warning Signal	Likelihood / Impact	Preventive Action	Owner
Technical	Cross-browser CSS incompatibility breaking form layout on older browsers.	Broken layout rendering during manual Firefox/Safari testing.	Medium / Medium	Use simple, standardized CSS flexbox layout without experimental features.	Cyril Costes
Teamwork	Unclear ownership of form validation logic leading to duplicate/conflicting code.	Two PRs modifying the same script.js file simultaneously.	High / Medium	Clearly isolate DOM layout tasks from JS validation script tasks in backlog issues.	Michael Girgis
Maintenance	Ad-hoc static UI tweak requests interrupting core sprint deliverables.	Frequent minor styling requests reported during daily updates.	Medium / Low	Establish a rotating weekly 'Support Guardian' role to handle ad-hoc tweaks.	Matthew Halliburton

Part F — Engineering Performance Dashboard

Metric Name	What it Measures	Why it is Useful	What it does NOT prove
PR Review Turnaround	Time from PR opening to merge.	Highlights review bottlenecks and pipeline flow.	Code quality or thoroughness of review.
First-Time-Right (FTR)	% of UI features passing initial PO verification without rework.	Measures effectiveness of acceptance criteria and local testing.	End-user satisfaction or business utility.
WIP Compliance	% of days within active WIP limit (<= 1).	Indicates adherence to single-tasking discipline.	Individual developer output volume.

Part G — Coaching & Leadership

1. Coaching Session Summary

Category	Detail
Context	Cyril struggled with writing the pure JavaScript logic to disable the submit button until all required form fields are valid.
Coaching Questions Asked	1. Which DOM events fire when a user changes input values? 2. How can you check if the dropdown selection is non-empty? 3. How can we test this step-by-step using browser console logs?
What Was Learned	Cyril learned to break down multi-field DOM validation into individual input event handlers rather than checking everything only on submit.
Action & Evaluation	Cyril writes a small event listener script; evaluated via code review during the next async check-in.

2. Decision Governance

Decision Type	Topic & Rationale
Collaborative Decision	Choosing the raw CSS approach (Flexbox vs. CSS Grid) for form layout. (Justified by team familiarity and daily developer workflow).
Manager Direction	Enforcing the WIP limit and review policies during sprint crunches. (Justified by the need to maintain pipeline stability).

Part H — Improvement Experiment

Experiment Parameter	Detail
Observed Problem	Pull requests lingered for >36 hours in code review.
Proposed Change	Review-First Policy: Review open PRs before pulling new tasks from the ready column.
Expected Result	PR turnaround time drops below 12 hours.
Measurement	Tracked via GitHub project pull request lead times over 5 days.
Result	Average turnaround dropped to 8.5 hours; team throughput increased by 25%.
Final Decision	Adopt (Permanently integrate into workflow agreement).

Code from the gemini so we can each just commit our parts in github:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>MediLink - Simple Client Intake</title>
    <style>
        /* Base Styles */
        * {
            box-sizing: border-box;
            margin: 0;
            padding: 0;
```

```
        font-family: Arial, sans-serif;
    }

    body {
        background-color: #f4f6f8;
        color: #333333;
        display: flex;
        flex-direction: column;
        min-height: 100vh;
    }

    /* Header */
    header {
        background-color: #0077b6;
        color: #ffffff;
        padding: 16px 20px;
        text-align: center;
    }

    header h1 {
        font-size: 24px;
        font-weight: bold;
    }

    header p {
        font-size: 13px;
        opacity: 0.9;
        margin-top: 4px;
    }

    /* Container & Card Layout */
    main {
        flex: 1;
        display: flex;
        justify-content: center;
        align-items: center;
        padding: 20px;
    }

    .card {
```

```
        background-color: #ffffff;
        border-radius: 8px;
        box-shadow: 0 2px 8px rgba(0, 0, 0, 0.1);
        width: 100%;
        max-width: 480px;
        padding: 24px;
    }

    .card h2 {
        font-size: 20px;
        color: #0077b6;
        margin-bottom: 16px;
        border-bottom: 2px solid #e0e0e0;
        padding-bottom: 8px;
    }

    /* Form Controls */
    .form-group {
        margin-bottom: 16px;
    }

    label {
        display: block;
        font-size: 14px;
        font-weight: bold;
        margin-bottom: 6px;
        color: #444444;
    }

    input[type="number"], select {
        width: 100%;
        padding: 10px;
        border: 1px solid #ccc;
        border-radius: 4px;
        font-size: 14px;
        background-color: #fff;
    }

    input[type="number"]:focus, select:focus {
        outline: none;
```

```
        border-color: #0077b6;
    }

    /* Error Styles */
    .error-message {
        color: #d9534f;
        font-size: 12px;
        margin-top: 4px;
        display: none;
    }

    input.invalid, select.invalid {
        border-color: #d9534f;
        background-color: #fff2f2;
    }

    /* Buttons */
    button {
        width: 100%;
        background-color: #0077b6;
        color: white;
        border: none;
        padding: 12px;
        font-size: 16px;
        font-weight: bold;
        border-radius: 4px;
        cursor: pointer;
        margin-top: 8px;
    }

    button:hover {
        background-color: #005b8e;
    }

    .secondary-btn {
        background-color: #6c757d;
        margin-top: 12px;
    }

    .secondary-btn:hover {
```

```
            background-color: #5a6268;
        }

        /* Confirmation View */
        .hidden {
            display: none;
        }

        .summary-list {
            list-style: none;
            margin-bottom: 20px;
        }

        .summary-list li {
            padding: 8px 0;
            border-bottom: 1px solid #eee;
            font-size: 14px;
        }

        .summary-list strong {
            color: #555;
        }

        /* Footer */
        footer {
            text-align: center;
            padding: 12px;
            font-size: 12px;
            color: #777777;
        }
    </style>
</head>
<body>

    <!-- Header Section -->
    <header>
        <h1>MediLink</h1>
        <p>Client Intake Portal</p>
    </header>
```

```
    <!-- Main Content Area -->
    <main>
        <div class="card">

            <!-- Intake Form -->
            <form id="intakeForm" novalidate>
                <h2>Client Intake Form</h2>

                <!-- Institution Selection -->
                <div class="form-group">
                    <label for="institution">Institution</label>
                    <select id="institution">
                        <option value="">-- Select Clinic --</option>
                        <option value="Montreal Medical Clinic">Montreal
Medical Clinic</option>
                        <option value="Downtown Dental Center">Downtown
Dental Center</option>
                        <option value="Eastside Health Clinic">Eastside
Health Clinic</option>
                    </select>
                    <div id="institutionError"
class="error-message">Please select an institution.</div>
                </div>

                <!-- Age Field -->
                <div class="form-group">
                    <label for="age">Age</label>
                    <input type="number" id="age" min="1"
placeholder="Enter age">
                    <div id="ageError" class="error-message">Please enter
a positive number for age.</div>
                </div>

                <!-- Gender Field -->
                <div class="form-group">
                    <label for="gender">Gender</label>
                    <select id="gender">
                        <option value="">-- Select Gender --</option>
                        <option value="Female">Female</option>
                        <option value="Male">Male</option>
```

```
                        <option value="Other">Other</option>
                        <option value="Prefer not to say">Prefer not to
say</option>
                    </select>
                    <div id="genderError" class="error-message">Please
select a gender option.</div>
                </div>

                <!-- Weight Field -->
                <div class="form-group">
                    <label for="weight">Weight (kg)</label>
                    <input type="number" id="weight" min="1"
placeholder="Enter weight in kg">
                    <div id="weightError" class="error-message">Please
enter a valid weight (> 0 kg).</div>
                </div>

                <!-- Height Field -->
                <div class="form-group">
                    <label for="height">Height (cm)</label>
                    <input type="number" id="height" min="1"
placeholder="Enter height in cm">
                    <div id="heightError" class="error-message">Please
enter a valid height (> 0 cm).</div>
                </div>

                <button type="submit">Submit Information</button>
            </form>

            <!-- Success Confirmation Screen -->
            <div id="confirmationScreen" class="hidden">
                <h2>Form Submitted Successfully</h2>
                <ul class="summary-list">
                    <li><strong>Institution:</strong> <span
id="sumInstitution"></span></li>
                    <li><strong>Age:</strong> <span
id="sumAge"></span></li>
                    <li><strong>Gender:</strong> <span
id="sumGender"></span></li>
```

```
                    <li><strong>Weight:</strong> <span
id="sumWeight"></span> kg</li>
                    <li><strong>Height:</strong> <span
id="sumHeight"></span> cm</li>
                </ul>
                <button type="button" class="secondary-btn"
id="editBtn">Edit Details</button>
            </div>

        </div>
    </main>

    <!-- Footer -->
    <footer>
        &copy; MediLink Appointment Management Platform
    </footer>

    <!-- JavaScript Logic -->
    <script>
        // DOM Elements
        const form = document.getElementById('intakeForm');
        const confirmationScreen =
document.getElementById('confirmationScreen');
        const editBtn = document.getElementById('editBtn');

        const fields = {
            institution: document.getElementById('institution'),
            age: document.getElementById('age'),
            gender: document.getElementById('gender'),
            weight: document.getElementById('weight'),
            height: document.getElementById('height')
        };

        const errors = {
            institution: document.getElementById('institutionError'),
            age: document.getElementById('ageError'),
            gender: document.getElementById('genderError'),
            weight: document.getElementById('weightError'),
            height: document.getElementById('heightError')
        };
```

```
// Form Submission Event Handler
form.addEventListener('submit', function(e) {
    e.preventDefault();

    let isValid = true;

    // Validate Institution
    if (fields.institution.value === "") {
        showError('institution');
        isValid = false;
    } else {
        hideError('institution');
    }

    // Validate Age (> 0)
    const ageVal = parseFloat(fields.age.value);
    if (isNaN(ageVal) || ageVal <= 0) {
        showError('age');
        isValid = false;
    } else {
        hideError('age');
    }

    // Validate Gender
    if (fields.gender.value === "") {
        showError('gender');
        isValid = false;
    } else {
        hideError('gender');
    }

    // Validate Weight (> 0)
    const weightVal = parseFloat(fields.weight.value);
    if (isNaN(weightVal) || weightVal <= 0) {
        showError('weight');
        isValid = false;
    } else {
        hideError('weight');
    }
```

```
            // Validate Height (> 0)
            const heightVal = parseFloat(fields.height.value);
            if (isNaN(heightVal) || heightVal <= 0) {
                showError('height');
                isValid = false;
            } else {
                hideError('height');
            }

            // If all fields are valid, display confirmation summary
            if (isValid) {
                document.getElementById('sumInstitution').textContent =
fields.institution.value;
                document.getElementById('sumAge').textContent =
fields.age.value;
                document.getElementById('sumGender').textContent =
fields.gender.value;
                document.getElementById('sumWeight').textContent =
fields.weight.value;
                document.getElementById('sumHeight').textContent =
fields.height.value;

                form.classList.add('hidden');
                confirmationScreen.classList.remove('hidden');
            }
        });

        // Helper function to display errors
        function showError(fieldName) {
            fields[fieldName].classList.add('invalid');
            errors[fieldName].style.display = 'block';
        }

        // Helper function to hide errors
        function hideError(fieldName) {
            fields[fieldName].classList.remove('invalid');
            errors[fieldName].style.display = 'none';
        }
```

```
        // Allow editing details from confirmation view
        editBtn.addEventListener('click', function() {
            confirmationScreen.classList.add('hidden');
            form.classList.remove('hidden');
        });
    </script>
</body>
</html>
```

Current Product:

